

Daniel Felipe Vargas Ulloa

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Double-degree last year engineering student at École Centrale Nantes and University of the Andes with industry experience in data analysis, machine learning, and software development. Seeking a 6-month final year internship from April to September 2027.

EDUCATION

École Centrale de Nantes

Diplôme d'Ingénieur (Master's level Engineering Degree)

Nantes, France

2025 – 2027

- Specialization in Computer Science for Information Systems

University of the Andes

Bachelor's in Systems and Computer Engineering

Minor in Computational Mathematics

Bogotá, Colombia

2021 – 2025

GPA: 4.4 / 5.0

- National Recognition among Top-Performing Students (Colombian Ministry of Education, 2024)

EXPERIENCE

Open-Source Python Library for Affective Data Analysis

École Centrale de Nantes

Sep 2026 – Present

Nantes, France

- Developed a modular, open-source Python library enabling cognitive science researchers to quantify human emotional responses using physiological and behavioral data.
- Designed data pipelines to quantify emotional responses and variability, validating reliability against public datasets.

Climate Risk Assessment of Colombia's Hydroelectric Infrastructure

University Jorge Tadeo Lozano

May 2026 – Sep 2026

Bogotá, Colombia

- Processed and analyzed NASA satellite observations and climate data to assess the vulnerability of Colombia's hydroelectric infrastructure to hydroclimatic stress.
- Developed quantitative indices to assess reservoir vulnerability and quantify the impact of El Niño-related drought on Colombia's hydroelectric system.

Clinical Anomaly Detection System for Pediatric Intensive Care Units Project

Central Military Hospital of Bogotá

Jan 2025 – Jul 2025

Bogotá, Colombia

- Developed a real-time predictive monitoring system for pediatric vital signs, reducing the false positive rate by 38.7%.
- Built end-to-end pipelines for data ingestion, signal processing, and detection of early clinical deterioration to support early intervention strategies.

Undergraduate Teaching Assistant, Software Development

University of the Andes

Jan 2024 – Jul 2024

Bogotá, Colombia

- Co-delivered a production oriented course on designing, implementing, and deploying full-stack applications using React and Spring microservices.
- Guided development teams on agile methodologies, version control, test-driven development and automated deployment.

SKILLS

Programming Languages: Python, SQL, Java, C/C++, JavaScript, TypeScript, Bash

Data & Machine Learning: Signal Processing, Predictive Modeling, Anomaly Detection, Data Pipelines, Power BI

Backend & Data Systems: Spring Boot, Flask, REST APIs, Docker, PostgreSQL, MongoDB, Linux, Git, Google Cloud

LANGUAGES

Spanish – Native · **English** – C2 (Fluent) · **French** – B2 (Advanced)